

Product	Ask Data
Category	AI + Data Analytics + Access Governance
Target Users	Admins, analysts, operations teams, business users
Core Value	Ask questions in plain English over approved data sources
Internal Store	MongoDB Atlas
Supported Sources	MySQL, PostgreSQL, MongoDB

1. Executive Summary

Ask Data is an AI-powered analytics workspace designed to help organizations unlock insights from approved databases without requiring every user to write SQL or understand database internals. Administrators control what is connected, who can access it, and what each employee is allowed to do. Employees then ask business questions in natural language, review generated results, visualize outcomes, and export them for decision-making.

The product addresses a very practical gap in modern teams: data exists, but access to answers is slow. Ask Data compresses the path from question to insight while preserving governance, usability, and operational clarity.

2. Problem Statement

- Business users depend on technical teams for simple data questions.
- Useful data is fragmented across SQL and NoSQL systems.
- Direct database access creates security and governance risks.
- Teams need a faster path from business question to report-ready output.

3. Solution Overview

- Admin-managed onboarding of trusted external data sources.
- Role-based permissions for each employee and connection.
- Natural-language analytics workflow backed by AI-assisted query generation.
- Schema discovery, summaries, charts, and export options in one interface.
- MongoDB Atlas as the single internal persistence store for the platform.

4. End-to-End Workflow

Step 1: Workspace Setup

An administrator registers, signs in, and becomes the control point for governed analytics access.

Step 2: Team Onboarding

The administrator creates employee accounts and assigns access permissions per connection.

Step 3: Data Source Integration

External MySQL, PostgreSQL, or MongoDB sources are configured and tested securely.

Step 4: Metadata and Schema Catalog

The platform extracts schema information so users can understand what data exists before querying it.

Step 5: Natural-Language Analysis

A user selects an approved source, asks a question in plain English, and receives generated results with summaries.

Step 6: Visualization and Export

The user turns results into charts and exports them to CSV, Excel, or PDF for downstream reporting.

5. Architecture

Layer	Implementation
Frontend	Next.js dashboards for admin and employee workflows
Backend	FastAPI service exposing auth, admin, employee, metadata, analysis, and export APIs
Internal Store	MongoDB Atlas collections for users, permissions, connections, metadata, logs, and counts
External Sources	MySQL, PostgreSQL, MongoDB / MongoDB Atlas
AI Flow	Groq/OpenAI-compatible client path with deterministic fallback behavior

6. Technical Highlights

- AI-assisted query generation with resilient fallback when external providers fail.
- JWT-based authentication and role-based authorization.
- Per-connection access control for governed usage.
- Encrypted storage of sensitive connection secrets.
- Metadata extraction for schema browsing and analytical context.
- Result visualization and multi-format export support.

7. Business and Hackathon Value

Ask Data stands out as a hackathon submission because it is not just an AI demo. It is a full product workflow that begins with governed onboarding and ends with usable insight delivery. The implementation combines authentication, permissions, connection management, schema intelligence, analytics, export, and a polished user experience into one coherent system.

- Solves a real enterprise productivity problem.
- Demonstrates strong full-stack engineering execution.
- Balances AI capability with governance and fallback safety.
- Has a clear path to real-world adoption and extension.

8. Current Feature Set

- Admin and employee authentication flows.
- Employee provisioning and permission assignment.

- Connection onboarding and validation for external data sources.
- Schema catalog and metadata browsing.
- Natural-language analytics over approved sources.
- Charts, summaries, and exported results.
- Recent query log visibility for admins.

9. Future Scope

Product Expansion

- Saved dashboards and reusable report templates.
- Scheduled report delivery and workspace notifications.
- Shared team collaboration around analytical outputs.

AI Improvements

- Smarter multi-source query planning and schema targeting.
- Explainability for table or collection selection.
- Cost-aware AI routing, caching, and optimization.

Governance and Enterprise Readiness

- Audit trails for permissions and query activity.
- Row-level access rules and approval workflows.
- Advanced observability and operational governance controls.

Performance and Deployment

- Background metadata refresh and connection health checks.
- Query caching and faster repeated analysis paths.
- Production deployment hardening, monitoring, and multi-tenant readiness.

10. Prize Positioning

For a 30K prize hackathon, Ask Data presents a strong case: it solves a high-value problem, demonstrates substantial technical depth, provides a visible user experience, and shows a credible roadmap beyond the prototype stage. The project combines product thinking with implementable engineering, which makes it compelling both as a demo and as a venture-ready concept.

11. Closing Note

Ask Data is a governed AI analytics platform built to make data access faster, safer, and more usable. It turns the distance between business questions and trustworthy answers into a structured product experience. That combination of relevance, execution, and future potential is what makes it a strong hackathon contender.